

Quantum Electronics & Qubits

Solution to Exercise Sheet 1 - Exercise 1

Prof. Dr. E. Bocquillon (bocquillon@ph2.uni-koeln.de)

Date of issue: 20/05/2025

Tutor: Dr. O. Atanov (atanov@ph2.uni-koeln.de)

Discussion: N/A

Classical Hall effect from Boltzmann equation

1. $f(\vec{r}, \vec{k}, t)$ is the distribution function that represents the probability of finding a particle with Bloch momentum $\vec{k}$ at position $\vec{r}$ at time t .
2. Starting from the relaxation-time approximation of the Boltzmann equation as given in the lecture:

$$\frac{\partial f}{\partial t} + \vec{v}_k \cdot \vec{\nabla}_r f + \frac{\vec{F}}{\hbar} \cdot \vec{\nabla}_k f = \frac{-f_1}{\tau} \quad (1)$$

where $f = f_0 + f_1$ is the perturbed distribution function and $\vec{F} = q(\vec{E} + \vec{v}_k \times \vec{B})$ is the total force acting on the particle with charge q . For a homogeneous system with time-independent distribution function that we are assuming here, the first two terms on the LHS of Eq. (1) become zero, leading to

$$\frac{q}{\hbar}(\vec{E} + \vec{v}_k \times \vec{B}) \cdot \vec{\nabla}_k(f_0 + f_1) = \frac{-f_1}{\tau} \quad (2)$$

$$\frac{q}{\hbar}\vec{E} \cdot (\hbar\vec{v}_k \frac{\partial f_0}{\partial \epsilon}) + \frac{q}{\hbar}\vec{E} \cdot \vec{\nabla}_k f_1 + \frac{q}{\hbar}(\vec{v}_k \times \vec{B}) \cdot (\hbar\vec{v}_k \frac{\partial f_0}{\partial \epsilon} + f_1) = \frac{-f_1}{\tau} \quad (3)$$

where in the last step, we used the relation $\vec{\nabla}_k f_0 = \hbar\vec{v}_k \frac{\partial f_0}{\partial \epsilon}$. Simplifying this equation further by noticing $(\vec{v}_k \times \vec{B}) \cdot \vec{v}_k = 0$, we get the expression:

$$q\vec{E} \cdot \vec{v}_k \left(\frac{\partial f_0}{\partial \epsilon}\right)\tau + \frac{q}{\hbar}\vec{E} \cdot \vec{\nabla}_k f_1 = -f_1 - \frac{q\tau}{\hbar}(\vec{v}_k \times \vec{B}) \cdot \vec{\nabla}_k f_1 \quad (4)$$

In the lecture, we have neglected the term $\vec{\nabla}_k f_1$ to keep the changes to 0th order only. We were allowed to do this because the transport is linear under an electric field. However, for the $\vec{B}$ term we have to keep this first-order contribution. With this, we finally arrive at the following:

$$q\vec{E} \cdot \vec{v}_k \left(\frac{\partial f_0}{\partial \epsilon}\right)\tau = -f_1 - \frac{q\tau}{\hbar}(\vec{v}_k \times \vec{B}) \cdot \vec{\nabla}_k f_1 \quad (5)$$

3. Let's now write the current density:

$$\vec{j}_e = 2q \int \frac{d^3k}{(2\pi)^3} \vec{v}_k f(\vec{r}, \vec{k}) = 2q \int \frac{d^3k}{(2\pi)^3} \vec{v}_k f_1(\vec{r}, \vec{k}) \quad (6)$$

since there is no current for the equilibrium distribution f_0 . Inserting f_1 from Eq. (5) and writing the current density vector in the component form using Einstein notation we find:

$$j_e^i = 2q^2 \int \frac{d^3k}{(2\pi)^3} v_k^i \left(-\frac{\partial f_0}{\partial \epsilon}\right) v_k^j \tau E^j - \frac{2q^2}{\hbar} \int \frac{d^3k}{(2\pi)^3} \tau v_k^i (\vec{v}_k \times \vec{B})^j \frac{\partial f_1}{\partial k^j} \quad (7)$$

which can be rearranged to give:

$$j_e^i = \underbrace{\left[2q^2 \int \frac{d^3k}{(2\pi)^3} v_k^i v_k^j \tau \left(-\frac{\partial f_0}{\partial \epsilon} \right) \right]}_{\sigma \delta_{ij}} E^j - \frac{2q^2}{\hbar} \int \frac{d^3k}{(2\pi)^3} \tau v_k^i (\vec{v}_k \times \vec{B})^j \frac{\partial f_1}{\partial k^j} \quad (8)$$

The first term is familiar to you from the lecture where we derived the Drude conductivity. The second term can be integrated by parts by noting that the τ is only energy dependent. Remember the formula $\int u dv = uv - \int v du$:

$$\begin{aligned} -\frac{2q^2}{\hbar} \int \frac{d^3k}{(2\pi)^3} \underbrace{\tau v_k^i (\vec{v}_k \times \vec{B})^j}_u \underbrace{\frac{\partial f_1}{\partial k^j}}_{dv} &= 0 + \frac{2q^2}{\hbar} \int \frac{d^3k}{(2\pi)^3} \left[\frac{\partial}{\partial k^j} [\tau v_k^i (\vec{v}_k \times \vec{B})^j] \right] f_1 \\ &= \frac{2q^2}{\hbar} \int \frac{d^3k}{(2\pi)^3} \frac{\partial \tau}{\partial \epsilon} \hbar v_k^j \underbrace{[v_k^i (\vec{v}_k \times \vec{B})^j]}_0 f_1 + \frac{2q^2}{\hbar} \int \frac{d^3k}{(2\pi)^3} \tau \frac{\partial}{\partial k^j} [v_k^i (\vec{v}_k \times \vec{B})^j] f_1 \end{aligned} \quad (9)$$

where the term $uv = 0$ due to $\vec{v}_k \cdot (\vec{v}_k \times \vec{B}) = 0$. The final term can be expanded as:

$$\begin{aligned} \frac{2q^2}{\hbar} \int \frac{d^3k}{(2\pi)^3} \tau \frac{\partial}{\partial k^j} [v_k^i (\vec{v}_k \times \vec{B})^j] f_1 &= \frac{2q^2}{\hbar} \int \frac{d^3k}{(2\pi)^3} \tau \frac{\partial v_k^i}{\partial k^j} (\vec{v}_k \times \vec{B})^j f_1 \\ &\quad + \frac{2q^2}{\hbar} \int \frac{d^3k}{(2\pi)^3} \tau v_k^j \underbrace{\frac{\partial}{\partial k^j} (\vec{v}_k \times \vec{B})^j}_0 f_1 \end{aligned} \quad (10)$$

where we used the relation:

$$\frac{\partial}{\partial k^j} (\vec{v}_k \times \vec{B})^j = \vec{\nabla} \cdot (\vec{v}_k \times \vec{B}) = \vec{B} \cdot (\vec{\nabla} \times \epsilon) = 0 \quad (11)$$

because the curl of a gradient is zero. Considering $\vec{v}_k = \frac{\hbar \vec{k}}{m^*}$, final form of the current density can be written as:

$$\begin{aligned} \vec{j}_e &= \sigma \vec{E} + \frac{2q^2}{\hbar} \int \frac{d^3k}{(2\pi)^3} \tau \frac{\partial v_k^i}{\partial k^j} (\vec{v}_k \times \vec{B})^j f_1 \\ &= \sigma \vec{E} + \frac{2q^2}{\hbar} \int \frac{d^3k}{(2\pi)^3} \tau \frac{\hbar}{m^*} (\vec{v}_k \times \vec{B})^i f_1 \end{aligned} \quad (12)$$

$$\vec{j}_e = \sigma \vec{E} + \frac{q\tau_F}{m^*} \vec{j}_e \times \vec{B} \quad (13)$$

4. Equation (13) can be rearranged to get the electric field:

$$\vec{E} = \frac{1}{\sigma} \vec{j}_e - \frac{q\tau_F}{m^*\sigma} \vec{j}_e \times \vec{B} \quad (14)$$

Inserting $\vec{B} = B\hat{k}$, $\vec{j}_e = j_e\hat{i}$ for our Hall effect geometry, we get:

$$\begin{aligned} E_x &= \frac{1}{\sigma} j_x + \frac{q\tau_F}{m^*\sigma} B j_y \\ E_y &= \frac{1}{\sigma} j_y - \frac{q\tau_F}{m^*\sigma} B j_x \end{aligned} \quad (15)$$

5. Noting the relation,

$$\begin{bmatrix} E_x \\ E_y \end{bmatrix} = \begin{bmatrix} \rho_{xx} & \rho_{xy} \\ \rho_{yx} & \rho_{yy} \end{bmatrix} \begin{bmatrix} j_x \\ j_y \end{bmatrix} \quad (16)$$

we can extract the relevant resistivity components:

$$\begin{aligned}\rho_{xx} = \rho_{yy} &= \frac{1}{\sigma} \\ \rho_{xy} = -\rho_{yx} &= -\frac{B}{nq}\end{aligned}\tag{17}$$

where we used the Drude conductivity formula $\sigma = \frac{nq^2}{m^*}\tau_F$ to simplify the last expression.

EINSTEIN SUMMATION NOTATION

Overview

In class, we began the discussion of how we can write vectors in a more convenient and compact convention. In addition to the advantage of compactness, writing vectors in this way allows us to manipulate vector calculations and prove vector identities in a much more elegant and less laborious manner.

The notation convention we will use, the Einstein summation notation, tells us that whenever we have an expression with a repeated index, we implicitly know to sum over that index from 1 to 3, (or from 1 to N where N is the dimensionality of the space we are investigating).

Vectors in Component Form

You are familiar with writing vectors in component form; the three dimensional vectors **A** and **B** can be expressed as:

$$\begin{aligned}\mathbf{A} &= A_x \hat{x} + A_y \hat{y} + A_z \hat{z} \\ \mathbf{B} &= B_x \hat{x} + B_y \hat{y} + B_z \hat{z}\end{aligned}$$

The dot product of the vectors, **A** and **B**, is:

$$\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z \quad (1)$$

We see immediately that the result of a dot product is a scalar, and that this resulting scalar is the sum of products. Since the dot product is a sum, we can write this as :

$$\mathbf{A} \cdot \mathbf{B} = \sum_{i=1}^3 A_i B_i \quad (2)$$

Where i is the arbitrary choice for indexing, and the summation runs from 1 to 3 to capture each of the three components of our vectors.

We can also write the expression in (2) in Einstein summation notation; since we do have a repeated index (in this case the index i), and our expression for a dot product becomes, simply:

$$\mathbf{A} \cdot \mathbf{B} = A_i B_i \quad (3)$$

where summation over i from 1 to 3 is assumed. Learning to write vectors in this notation will make our later work enormously easier.

Our First Vector Proof

Let's consider our first proof, and we will solve it using standard component notation and then solve it using Einstein summation notation. We will show that :

$$\frac{d}{dt} (\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \cdot \frac{d\mathbf{B}}{dt} + \mathbf{B} \cdot \frac{d\mathbf{A}}{dt} \quad (4)$$

where the vectors $\mathbf{A}$ and $\mathbf{B}$ are both functions of time. Using component notation, we write out the dot product of $\mathbf{A}$ and $\mathbf{B}$ using (1) from above :

$$\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z$$

taking the derivative, and using the product rule for differentiation :

$$\begin{aligned} \frac{d}{dt} (\mathbf{A} \cdot \mathbf{B}) &= \frac{d}{dt} (A_x B_x + A_y B_y + A_z B_z) = \\ &A_x \frac{dB_x}{dt} + B_x \frac{dA_x}{dt} + A_y \frac{dB_y}{dt} + B_y \frac{dA_y}{dt} + A_z \frac{dB_z}{dt} + B_z \frac{dA_z}{dt} \end{aligned}$$

We can group these terms :

$$\left[A_x \frac{dB_x}{dt} + A_y \frac{dB_y}{dt} + A_z \frac{dB_z}{dt} \right] + \left[B_x \frac{dA_x}{dt} + B_y \frac{dA_y}{dt} + B_z \frac{dA_z}{dt} \right] \quad (5)$$

Look carefully at the terms in the brackets and compare with equation (1). Notice that the bracket on the left is just $\mathbf{A} \cdot \frac{d\mathbf{B}}{dt}$ and that the bracket on the right is $\mathbf{B} \cdot \frac{d\mathbf{A}}{dt}$, and we have proven our theory using component notation.

In Einstein summation notation, we begin with the definition of the dot product :

$$\mathbf{A} \cdot \mathbf{B} = A_i B_i$$

and differentiate using the product rule. Remembering that all of our terms on the right are scalars, we write :

$$\frac{d}{dt} (\mathbf{A} \cdot \mathbf{B}) = \frac{d}{dt} (A_i B_i) = A_i \frac{dB_i}{dt} + B_i \frac{dA_i}{dt} \quad (6)$$

The terms on the right each have a repeated index, so we recognize each term on the right represents a summation over three coordinates. Compare the terms on the right in (6) with the terms in (5), and you will see that the right hand terms in (6) are equivalent to the bracketed terms in (5); in other words, eq. (6) is a one line proof of our identity; all that remains is to equate this to $\frac{d}{dt} (\mathbf{A} \cdot \mathbf{B})$. This simple vector proof shows the power of using Einstein summation notation.

The general game plan in using Einstein notation summation in vector manipulations is:

- Write down your identity in standard vector notation;
- "Translate" the vectors into summation notation; this will allow you to work with the scalar components of the vectors;
- Manipulate the scalar components as needed;
- "Translate" the scalar result back into vector form.

In the proof above, point two is accomplished when you write:

$$\frac{d}{dt} (\mathbf{A} \cdot \mathbf{B}) = \frac{d}{dt} (A_i B_i),$$

point three is accomplished with the differentiation :

$$A_i \frac{dB_i}{dt} + B_i \frac{dA_i}{dt}$$

and the final result occurs on recognition that is the result we wish to prove.

Cross Products and Einstein Summation Notation

In class, we studied that the vector product between two vectors A and B is called the cross product and written as :

$$\mathbf{C} = \mathbf{A} \times \mathbf{B}$$

The resulting vector C has magnitude equal to $|A| |B| \sin\theta$, and has a direction mutually perpendicular to the vectors A and B . We showed that in component notation, the cross product is :

$$\mathbf{C} = (A_y B_z - A_z B_y) \hat{\mathbf{x}} + (A_z B_x - A_x B_z) \hat{\mathbf{y}} + (A_x B_y - A_y B_x) \hat{\mathbf{z}} \quad (7)$$

The components of the cross product reveal a number of patterns :

- Each component of the vector C is a difference of products.
- Each component has a permutation of terms, that is, each component consists of terms of the form :

$$A_\alpha B_\beta - A_\beta B_\alpha$$

- The $\hat{x}$ component of $\mathbf{C}$ has no terms involving A_x or B_x ; the same condition applies to the $\hat{y}$ and $\hat{z}$ components of $\mathbf{C}$.

Before we can write cross products in summation notation, we need to develop a mathematical formalism that will reproduce the patterns we describe above. Just such a formalism exists, and it is called the **Levi-Civita permutation tensor**, or just the permutation tensor. This permutation tensor can be written for any number of dimensions, but for the most part we will be dealing with three dimensional space.

The permutation tensor is written as ϵ_{ijk} where i, j , and k are indices corresponding to the three coordinate directions. The permutation tensor is defined to have the following values:

- 0 if any two indices are the same

$\epsilon_{ijk} = +1$ if all three indices are different and are cyclic

- -1 if all three indices are different and are anti - cyclic

Cyclic permutations (or even permutations) are 123, 231, and 312; anti - cyclic (or odd) permutations are 132, 213, and 321.

Properties of the Permutation Tensor

Before moving on to how we write cross products with the permutation tensor, let's investigate some of its properties.

First, how many different ways can we write the tensor in 3 dimensions? The permutation tensor has three indices, ϵ_{ijk} , and there are three ways of choosing i (it could be 1, 2 or 3), three ways of choosing j , and three ways of choosing k . Therefore, there are 27 possible ways of writing ϵ_{ijk} in 3-space.

While tedious, it is instructive to write all of these possible values:

$$\begin{array}{lll}
 \epsilon_{111} & \epsilon_{211} & \epsilon_{311} \\
 \epsilon_{112} & \epsilon_{212} & \epsilon_{312} \\
 \epsilon_{113} & \epsilon_{213} & \epsilon_{313} \\
 \epsilon_{121} & \epsilon_{221} & \epsilon_{321} \\
 \epsilon_{122} & \epsilon_{222} & \epsilon_{322} \\
 \epsilon_{123} & \epsilon_{223} & \epsilon_{323} \\
 \epsilon_{131} & \epsilon_{231} & \epsilon_{331} \\
 \epsilon_{132} & \epsilon_{232} & \epsilon_{332} \\
 \epsilon_{133} & \epsilon_{233} & \epsilon_{333}
 \end{array}$$

Of these 27 different values, 21 of them are zero, since they have one or more indices the same (refer to the first of the rules for the value of the permutation tensor above). Of the remaining six possible values, three of them are cyclic permutations (noted in blue) and have value +1, and six are anti-cyclic permutations of value -1 (noted in red).

Writing Cross Products with the Permutation Tensor

Let's start by rewriting eq. (7) from above :

$$\mathbf{C} = (A_y B_z - A_z B_y) \hat{\mathbf{x}} + (A_z B_x - A_x B_z) \hat{\mathbf{y}} + (A_x B_y - A_y B_x) \hat{\mathbf{z}}$$

We will show that any component of the vector $\mathbf{C}$, produced from the cross product of vectors $\mathbf{A}$ and $\mathbf{B}$, can be written in the form :

$$C_i = \epsilon_{ijk} A_j B_k \quad (8)$$

This means that the i^{th} component of $\mathbf{C}$ (where i can be 1, 2, or 3) can be expressed in terms of the permutation tensor. Let's go into some detail to see how this works. For specificity, let's say we want to find the x component of $\mathbf{C}$, in other words, we are going to set $i=1$ (since $i=1$ corresponds to the x component, $i=2$ corresponds to the y component, $i=3$ to the z component).

If we set $i=1$, then we know that the values of j and k must be 2 or 3, since the permutation tensor is zero if there are any repeated

indices. Looking at (8) above, we can see that both j and k are repeated on the right hand side, so we know we will sum over j and k . (Since we have set $i=1$ already, I will not show explicitly the summation for j or $k=1$). Thus, with $i=1$, our implied summation in (8) is:

$$C_1 = \epsilon_{122} A_2 B_2 + \epsilon_{123} A_2 B_3 + \epsilon_{132} A_3 B_2 + \epsilon_{133} A_3 B_3 \quad (9)$$

The first and last terms on the right hand side of (9) are zero because there are duplicated indices; the terms highlighted in blue and red are the only non zero terms. The term highlighted in blue has a value of $+1$ and the value highlighted in red has a value of -1 , so that equation (9) tells us that the x component of C is :

$$C_1 = A_2 B_3 - A_3 B_2 \quad (10)$$

Compare this to the form of eq. (7) and you will see this is exactly equivalent to the expression we obtained from a term by term multiplication of components.

What happens if we set $i=2$? We know that j and k will be either 1 or 3, and the only non zero terms that survive the summation are:

$$C_2 = \epsilon_{213} A_1 B_3 + \epsilon_{231} A_3 B_1 = A_3 B_1 - A_1 B_3 \quad (11)$$

Similarly, if $i = 3$, we find the z component of C as :

$$C_3 = \epsilon_{321} A_2 B_1 + \epsilon_{312} A_1 B_2 = A_1 B_2 - A_2 B_1 \quad (12)$$

And we have accurately reproduced each of the individual components of the vector C . A simple final step is to combine all these components to yield the complete vector, so we need to sum all the components as :

$$C_i e_i = \epsilon_{ijk} A_j B_k e_i \quad (13)$$

where we make our first use of " e_i " to mean the i^{th} unit vector; throughout the semester we will make use of $\hat{e}$ to refer to a unit vector in some direction; if we expand the left side of eq. (12) in accordance with our understanding of Einstein summation, we get:

$$C_i e_i = C_1 \hat{e}_1 + C_2 \hat{e}_2 + C_3 \hat{e}_3 = C_1 \hat{x} + C_2 \hat{y} + C_3 \hat{z} \quad (14)$$

As a quick summation, the i^{th} component of the cross product resulting from $\mathbf{C} = \mathbf{A} \times \mathbf{B}$ is:

$$C_i = \epsilon_{ijk} A_j B_k \quad (15)$$

and the complete vector C is :

$$C_i e_i = \epsilon_{ijk} A_j B_k e_i \quad (16)$$

You should not be confused by the slight differences between (15) and (16); (15) tells you the value of any individual component; (16) is the complete vector, and is computed by adding up all the individual components.

A Proof Using Dot and Cross Products

We know that the cross product of two vectors produces a third vector that is perpendicular to each of the original vectors. In other words, if $\mathbf{C} = \mathbf{A} \times \mathbf{B}$, we expect that $\mathbf{A} \cdot \mathbf{C}$ and $\mathbf{B} \cdot \mathbf{C}$ should be zero. You can probably imagine how many steps this takes in component form, let's prove this using summation notation. We know the i^{th} component of the vector $\mathbf{C}$ can be expressed as:

$$C_i = \epsilon_{ijk} A_j B_k \quad (17)$$

Taking the dot product of $\mathbf{A}$ and $\mathbf{C}$ is equivalent to finding $A_i C_i$, so we can write:

$$\mathbf{A} \cdot \mathbf{C} = A_i C_i = A_i \epsilon_{ijk} A_j B_k \quad (18)$$

All the terms on the right hand side of (18) are scalars, so we can reorder them in any way we wish since multiplication of scalars is commutative. In particular, we can rewrite (18) as :

$$A_i \epsilon_{ijk} A_j B_k = B_k \epsilon_{ijk} A_i A_j \quad (19)$$

The final term on the right is the k^{th} component of the cross product of $\mathbf{A} \times \mathbf{A}$. Since the cross product of any vector with itself is zero, we have shown that $\mathbf{A} \cdot \mathbf{C}$ is zero. An exactly similar analysis will show that $\mathbf{B} \cdot \mathbf{C}$ is zero.

Einstein summation convention and δ -functions

Contents

1	Summation convention	1
1.1	Summation convention basics	1
1.2	The Kronecker- δ and the ϵ -tensor	2
1.3	Summation convention and operators	3
2	δ-functions	4

1 Summation convention

1.1 Summation convention basics

Einstein summation convention is a convenient notation when manipulating expressions involving vectors, matrices, or tensors in general. (A tensor is a collection of numbers labeled by indices. The *rank* of a tensor is the number of indices required to specify an entry in the tensor, so a vector is a rank-1 tensor, whereas a matrix is a rank-2 tensor.)

The “rules” of summation convention are:

- Each index can appear at most twice in any term.
- Repeated indices are implicitly summed over.
- Each term must contain identical non-repeated indices.

For example,

$$M_{ij}v_j \equiv \sum_i M_{ij}v_j$$

is a valid expression (it is just left-multiplication of vector $\mathbf{v}$ by matrix M), whereas

$$M_{ij}u_jv_j + w_i$$

is not a valid expression in summation convention, since the index j appears three times in the first term, and

$$T_{ijk}u_k + M_{ip}$$

is invalid since the first term contains the non-repeated index j whereas the second term contains p .

The great advantage of using summation convention is that all quantities in an expression become scalars. And we are free to re-order scalars in whatever way we like (though care must be taken when an expression includes operators, see section 1.3).

1.2 The Kronecker- δ and the ϵ -tensor

When writing vector expressions in summation convention, two tensors appear so often that they are conventionally always given the same symbol: the Kronecker- δ and the ϵ -tensor (also called the Levi-Civita symbol, or the anti-symmetric tensor).

The Kronecker- δ is a rank-2 tensor, defined by:

$$\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}.$$

In an expression, it has the effect of replacing one index with another (remember the implicit summation):

$$\delta_{ij}u_j = u_i$$

The Kronecker- δ can be thought of as the identity matrix, e.g. in three dimensions

$$\delta_{ij} = \left[\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right]_{ij}.$$

The ϵ -tensor is rank-3, and is defined by:

$$\epsilon_{ijk} = \begin{cases} 1 & ijk \text{ even} \\ -1 & ijk \text{ odd} \\ 0 & \text{otherwise.} \end{cases}$$

The “even” and “odd” refer to the permutation defined by ijk (e.g. 123 is even whereas 132 is odd), and the “otherwise” includes all combinations with repeated indices. Note that the definition implies that we are always free to cyclically permute indices:

$$\epsilon_{ijk} = \epsilon_{kij} = \epsilon_{jki}.$$

On the other hand, swapping any two indices gives a sign-change:

$$\epsilon_{ijk} = -\epsilon_{ikj},$$

which explains why the ϵ -tensor is sometimes called the completely anti-symmetric tensor. The ϵ -tensor is very useful for writing the vector product (or “cross product”) in summation convention:

$$[\mathbf{u} \times \mathbf{v}]_i = \epsilon_{ijk}u_jv_k.$$

An important identity connects the ϵ -tensor with the Kronecker- δ :

$$\epsilon_{ijk}\epsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}.$$

1.3 Summation convention and operators

Vector operators are easily handled using summation convention. The vector operator ∇ can be thought of as a vector. E.g. in three dimensions

$$\nabla = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix}.$$

The grad, div and curl operations in summation convention look like:

$$\begin{aligned} [\text{grad } \phi]_i &= [\nabla \phi]_i = \nabla_i \phi, \\ [\text{div } \mathbf{A}]_i &= [\nabla \cdot \mathbf{A}]_i = \nabla_i A_i, \\ [\text{curl } \mathbf{A}]_i &= [\nabla \times \mathbf{A}]_i = \epsilon_{ijk} \nabla_j A_k. \end{aligned}$$

Note that all operators in summation convention become scalar operators.

The only thing to be careful of when dealing with operators in summation convention is that unlike scalars, scalar *operators* can not necessarily be freely re-ordered, since they must always appear to the left of whatever they operate on. For instance

$$\nabla_i A_i \neq A_i \nabla_i.$$

2 δ -functions

In section 1.2 we met the Kronecker- δ , which is zero unless it's two indices are identical. The indices of the Kronecker δ take discrete values (integers). The δ -function (also called the Dirac δ -function) is the continuous analogue: $\delta(x)$ is zero everywhere apart from the origin $x = 0$. The analogy is clearer if we write the function $\delta(x - y)$, which is zero everywhere except at $x = y$.

Formally, the δ -function is defined by

$$\int_{-\infty}^{+\infty} f(x)\delta(x-a)dx = f(a) \quad \forall f(x),$$

where $f(x)$ is any function.

It can also be expressed as the limit of certain functions. For example

$$\delta(x) = \lim_{\sigma \rightarrow 0} e^{-x^2/\sigma} = \lim_{k \rightarrow \infty} \frac{\sin(kx)}{kx}.$$

However, when proving fundamental properties of the δ -function, the definition is more useful than these explicit expressions.

The one-dimensional δ -function defined above can easily be extended to higher dimensions in the obvious way. Higher dimensional δ -functions are just products of one-dimensional ones. E.g. in three dimensions,

$$\delta(\mathbf{r} - \mathbf{r}_0) = \delta(x - x_0)\delta(y - y_0)\delta(z - z_0).$$

As already stated, the δ -function is zero everywhere except at the origin, where it diverges. However, as is obvious from the definition, the area under the δ function is finite:

$$\int_{-\infty}^{+\infty} \delta(x)dx = 1.$$

It has a number of other interesting (and important) properties. Since it is zero everywhere other than the origin, it is also true that

$$\int_p^q f(x)\delta(x-a)dx = f(a) \quad \text{for } p < a < q.$$

Rescaling its argument re-scales the δ -function itself:

$$\delta(ax) = \frac{1}{|a|}\delta(x).$$

The derivative of the δ -function is given by

$$\int_p^q dx f(x) \frac{d}{dx} \delta(x-a) = - \left. \frac{df(x)}{dx} \right|_{x=a}.$$

Finally, if $f(x)$ has simple zeros at $x = x_1, x_2, \dots, x_N$, then

$$\delta(f(x)) = \sum_{i=1}^N \frac{\delta(x - x_i)}{\left| \frac{df}{dx}(x_i) \right|}.$$

In mechanics, the δ -function $\delta(t-t_0)$ is used to describe an ideal unit impulse (a “kick”) at time $t = t_0$. The response function of a system is defined in terms of an ideal unit impulse. (Since there is an analogy between mechanical systems and electric circuits, it is no surprise that all this applies to circuits too). It is used in electromagnetism to describe idealized point-charges: $q\delta(\mathbf{r}-\mathbf{r}_0)$ describes a point-charge q situated at $\mathbf{r}_0$.